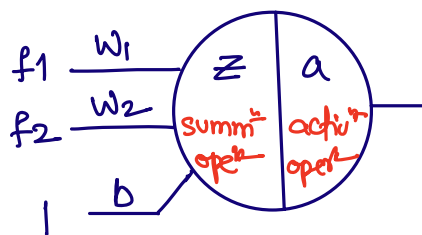


Perceptron



$$z = \left[\sum_{i=1}^n w_i f_i \right] + b$$

Activation operation:

goal of activation operation is to introduce non-linearity behavior

Ⓐ Pass through function

$$f(x) = x \dots \text{regression}$$

Ⓑ sigmoid function

$$f(x) = \frac{1}{1 + e^{-x}} \dots \text{binary classification}$$

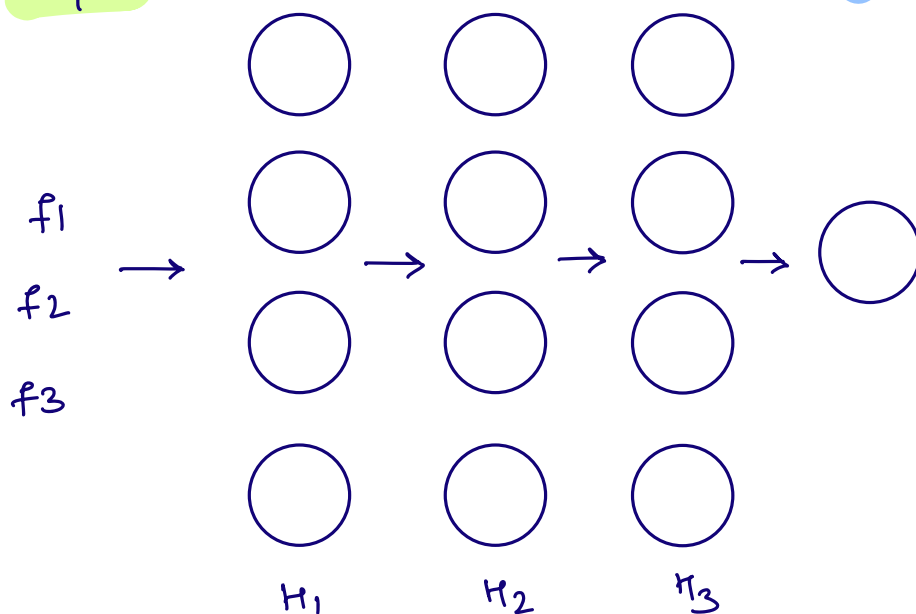
returns probability.

MLP (multi-layer perceptron) | ANN (Artificial Neural Network)

Input Layer

Set of Hidden Layers

output layer



Model Training Process

① Forward Propagation

Calc z

Calc a

② Back Propagation

Calc Error

Calc Gradients (Δw & Δb)

Update w & b

Back propagation by

Sir Geoffrey Hinton

Q Solve MLP 😊

Given dataset D containing one record

$D \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$ target $[1]$

f_1 f_2 f_3

$$\eta = \gamma = 0.9$$

assume the initialized weights and bias are,

w_{per} next

$$w_{14} = 0.2$$

$$w_{15} = -0.3$$

$$w_{24} = 0.4$$

$$w_{25} = 0.1$$

$$w_{34} = -0.5$$

$$w_{35} = 0.2$$

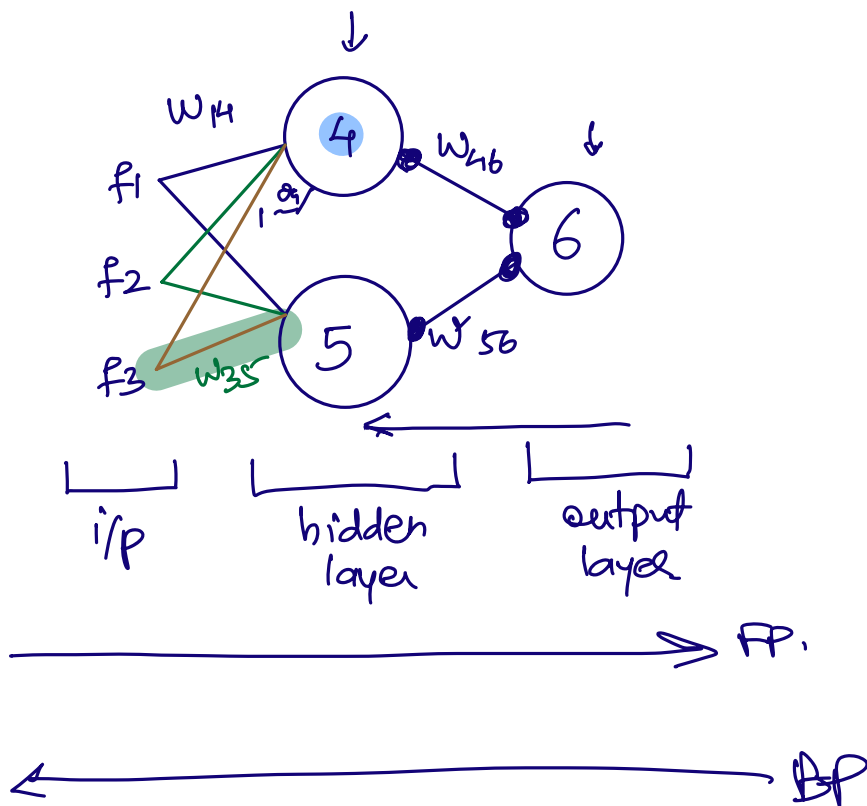
$$w_{46} = -0.3$$

$$w_{56} = -0.2$$

$$O_4 = -0.4$$

$$O_5 = 0.2$$

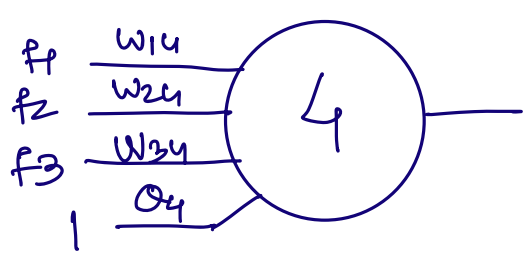
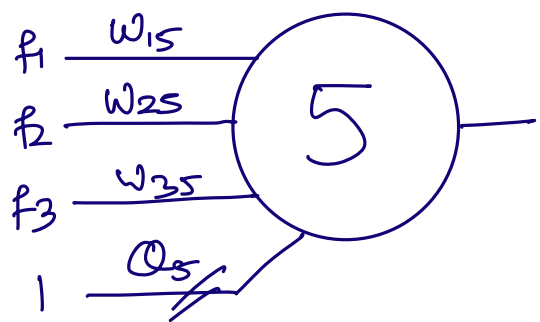
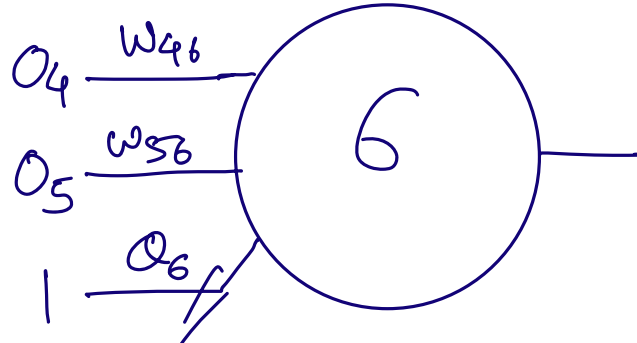
$$O_6 = 0.1$$



Solⁿ

Epoch = 1

Step 1: Forward Propagation

Node	z	a
4	 $z = [w_{14}f_1 + w_{24}f_2 + w_{34}f_3] + O_4(1)$ $= -0.7 //$	$O_4 = \frac{1}{1 + e^{+0.7}}$ $= \underline{0.3318} //$
5	 $z = 0.1$	$O_5 = \frac{1}{1 + e^{-0.1}}$ $= 0.5250$
6	 $z = -0.1045$	$O_6 = 0.4738$

Step 2 Backpropagation Starts

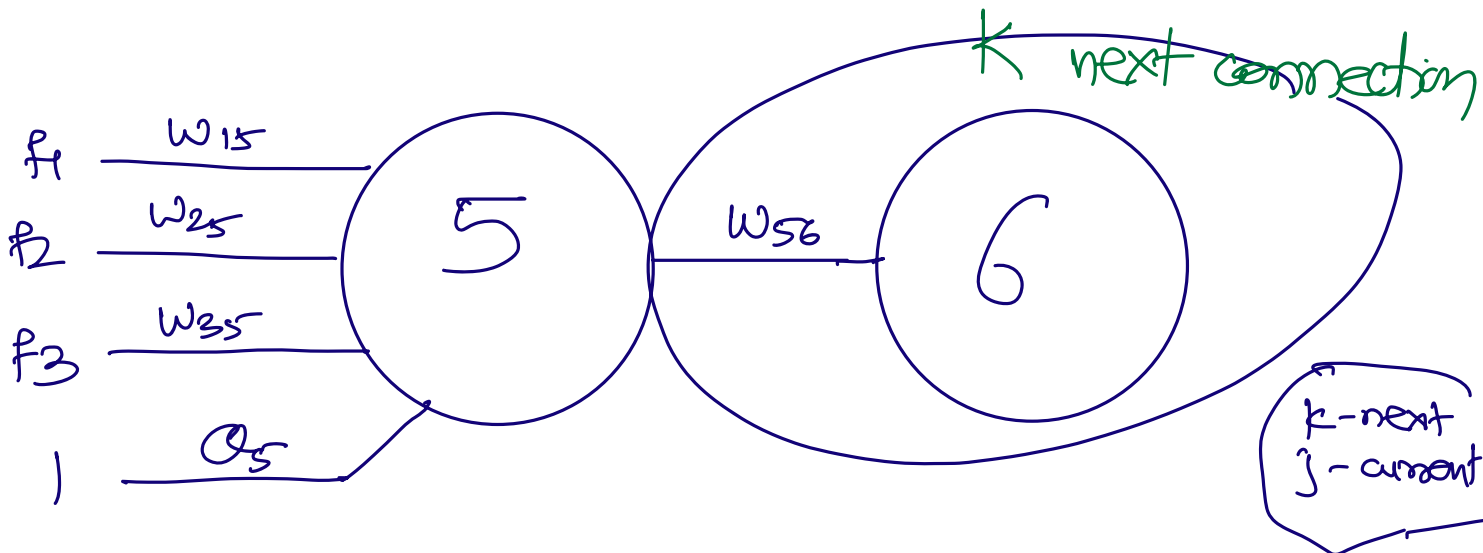
2a. Calc errors

- // Backpropagate the errors:
- for each unit j in the output layer
 - $Err_j = O_j(1 - O_j)(T_j - O_j)$; // compute the error
 - for each unit j in the hidden layers, from the last to the first hidden layer
 - $Err_j = O_j(1 - O_j) \sum_k Err_k w_{jk}$; // compute the error with respect to the next higher layer, k

for node 6

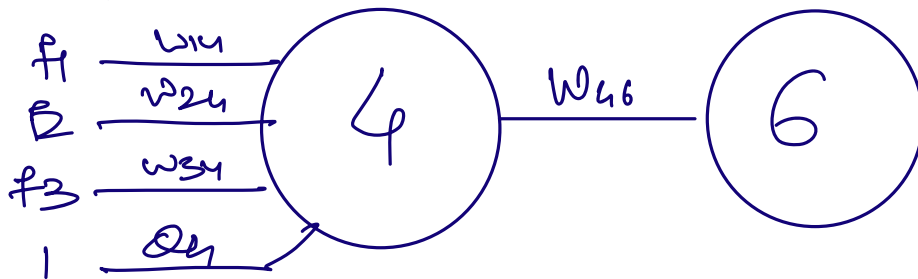
$$\begin{aligned} Err_6 &= O_6(1 - O_6)(T_6 - O_6) \\ &= 0.4738(1 - 0.4738)(1 - 0.4738) \\ &= \underline{\underline{0.1311}} \end{aligned}$$

for node 5



$$\begin{aligned} Err_j &= O_j(1 - O_j) \sum_k Err_k w_{jk} \\ &= O_5(1 - O_5) [Err_6 w_{56}] \\ &= 0.5250(1 - 0.5250) [(0.1311)(-0.2)] \\ &= -0.0065 \end{aligned}$$

for node 4



$$\begin{aligned} \text{Err}_4 &= O_4 (1 - O_4) [\text{Err}_6 w_{46}] \\ &= -0.0087 \end{aligned}$$

step 2 b. calc gradients

w_{ij}

$$\Delta w = \eta \cdot \text{Err}_j \cdot O_i$$

$$\begin{aligned} w_{46} &= w_{46} + [\eta \cdot \text{Err}_6 \cdot O_4] \\ &= -0.3 + [(0.9)(0.1311)(0.3318)] \\ &= -0.2609 \end{aligned}$$

$$\begin{aligned} O_4 &= O_4 + [(0.9)(-0.0087)] \\ &= -0.4080 \end{aligned}$$

$$\Delta O_j = \eta \cdot \text{Err}_j$$

Epoch 1

$$① W_{14} = 0.1921$$

$$9 Q_4 = -0.4079$$

$$② W_{24} = 0.4000$$

$$③ W_{34} = -0.5079$$

$$④ W_{15} = -0.3059$$

$$10 Q_5 = 0.1941$$

$$⑤ W_{25} = 0.1000$$

$$⑥ W_{35} = 0.1941$$

$$⑦ W_{46} = -0.2608$$

$$11 Q_6 = 0.2187$$

$$⑧ W_{56} = -0.1380$$

11

t.f. keras

Sequential

